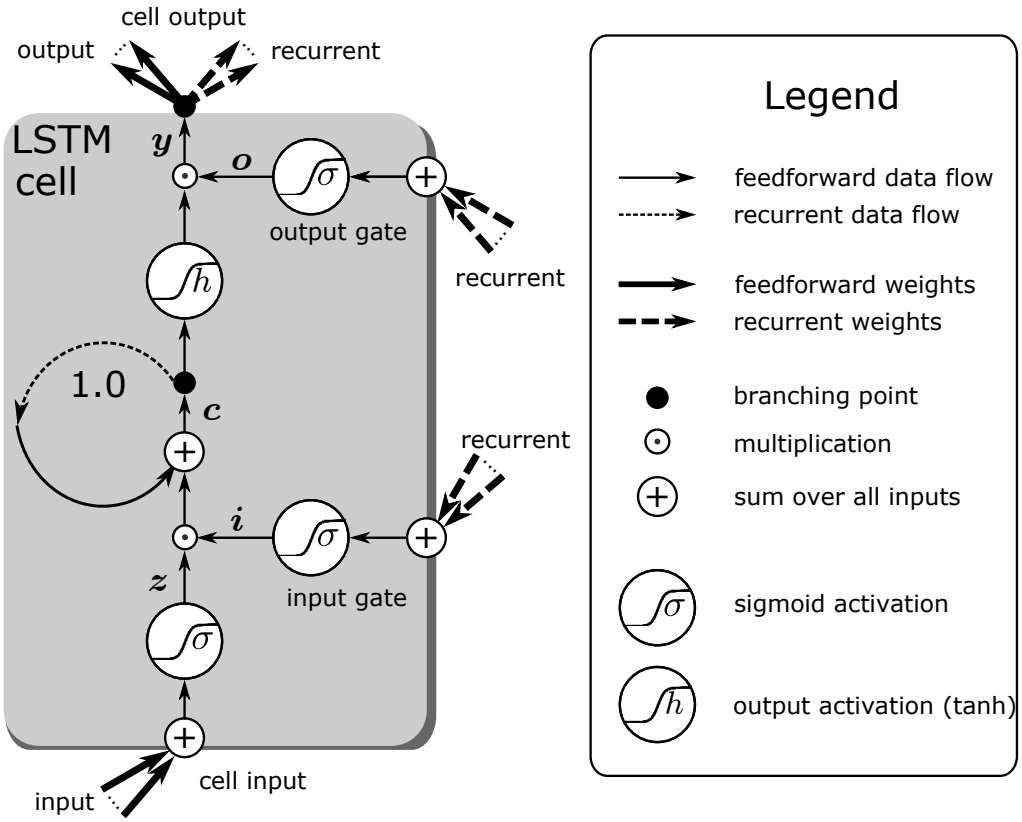

Figure A12: LSTM memory cell used for Layer-Wise Relevance Propagation (LRP). $\boldsymbol{z}$ is the vector of cell input activations, $\boldsymbol{i}$ is the vector of input gate activations, $\boldsymbol{c}$ is the vector of memory cell states, $\boldsymbol{o}$ is the vector of output gate activations, and $\boldsymbol{y}$ is the vector of cell output activations. The activation functions are the sigmoid $\sigma(x) = a_g \frac{1}{1+\exp(-x)}$ and the cell state activation $h(x) = a_h \tanh(x)$. Data flow is either "feed-forward" without delay or "recurrent" with an one-step delay. External input reaches the LSTM network only via the cell input $\boldsymbol{z}$. All gates only receive recurrent input, that is, from other memory cells.

Since we assume $c^0 = 0$, we have

$$c^T \;=\; \sum_{t=1}^{T} i^t \; z^t \tag{A351}$$

and therefore

$$R_{z^t} \;=\; \frac{i^t \; z^t}{\sum_{\tau=1}^{T} i^\tau \; z^\tau} \; R_{y^T} \; . \tag{A352}$$

Therefore the relevance R_{y^T} is distributed across the inputs z^t for $t = 1 \ldots T-1$, where input z^t obtains relevance R_{z^t}.

A8.4.3 LSTM for Nondecreasing Memory Cells

contribution analysis is made simpler if memory cells are nondecreasing since the contribution of each input to each memory cells is well defined. The problem that a negative and a positive input cancels each other is avoided. For nondecreasing memory cells and contribution analysis with LSTM, we make following assumptions:

(A1) $\boldsymbol{f}^t = 1$ for all t. That is the forget gate is always 1 and nothing is forgotten. We assume uniform credit assignment, which is ensured by the forget gate set to one.

(A2) $g > 0$, that is, g is positive. For example we can use a sigmoid $\sigma(x) = a_g \frac{1}{1+\exp(-x)}$: $g(x) = a_g \sigma(x)$, with $a_g = 2, 3, 4$. With a positive g all contributions are positive. The

cell input $\boldsymbol{z}$ (the function g) has a negative bias, that is, $\boldsymbol{b_z} < 0$. This is important to avoid the drift effect. The drift effect is that the memory content only gets positive contributions which lead to an increase of $\boldsymbol{c}$ over time. Typical values are $\boldsymbol{b_z} = -1, -2, -3, -4, -5$.

(A3) We want to ensure that $h(0) = 0$. If the memory content is zero then nothing is transferred to the next layer. Therefore we set $h = a_h \tanh$ with $a_h = 1, 2, 4$.

(A4) The cell input gate $\boldsymbol{z}$ is only connected to the input but not to other memory cells. $\boldsymbol{W_z}$ has only connections to the input.

(A5) The input gate $\boldsymbol{i}$ is not connected to the input, that is, $\boldsymbol{W_i}$ has only connections to other memory cells.

(A6) The output gate $\boldsymbol{o}$ is not connected to the input, that is, $\boldsymbol{W_o}$ has only connections to other memory cells.

(A7) The input gate $\boldsymbol{i}$ has a negative bias, that is, $\boldsymbol{b_i} < 0$. Like with the cell input the negative bias avoids the drift effect. Typical values are $\boldsymbol{b_i} = -1, -2, -3, -4$.

(A8) The output gate $\boldsymbol{o}$ may also have a negative bias, that is, $\boldsymbol{b_o} < 0$. This allows to bring in different memory cells at different time points. It is related to resource allocation.

(A9) The memory cell content is initialized with zero at time $t = 0$, that is, $\boldsymbol{c}^0 = 0$. We ensured via the architecture that $\boldsymbol{c}^t \geqslant 0$ and $\boldsymbol{c}^{t+1} \geqslant \boldsymbol{c}^t$, that is, the memory cells are positive and nondecreasing.

The resulting LSTM forward pass rules for nondecreasing memory cells are:

$$\boldsymbol{z}^t = a_g\, \sigma\left(\boldsymbol{W_z}\, \boldsymbol{x}^t + \boldsymbol{b_z}\right) \qquad \text{cell input} \tag{A353}$$

$$\boldsymbol{i}^t = \sigma\left(\boldsymbol{W_i}\, \boldsymbol{x}^t + \boldsymbol{b_i}\right) \qquad \text{input gate} \tag{A354}$$

$$\boldsymbol{c}^t = \boldsymbol{i}^t \odot \boldsymbol{z}^t + \boldsymbol{c}^{t-1} \qquad \text{cell state} \tag{A355}$$

$$\boldsymbol{o}^t = \sigma\left(\boldsymbol{W_o}\, \boldsymbol{x}^t + \boldsymbol{b_o}\right) \qquad \text{output gate} \tag{A356}$$

$$\boldsymbol{y}^t = \boldsymbol{o}^t \odot a_h \tanh\left(\boldsymbol{c}^t\right) \qquad \text{cell output} \tag{A357}$$

See Figure A13 for a LSTM memory cell that is nondecreasing.

A8.4.4 LSTM without Gates

The most simple LSTM architecture for contribution analysis does not use any gates. Therefore complex dynamics that have to be treated in the contribution analysis are avoided. For LSTM without gates, we make following assumptions:

(A1) $\boldsymbol{f}^t = 1$ for all t. That is the forget gate is always 1 and nothing is forgotten.

(A2) $\boldsymbol{o}^t = 1$ for all t. That is the output gate is always 1.

(A3) $\boldsymbol{i}^t = 1$ for all t. That is the input gate is always 1.

(A4) $g > 0$, that is, g is positive. For example we can use a sigmoid $\sigma(x) = a_g \frac{1}{1+\exp(-x)}$: $g(x) = a_g \sigma(x)$, with $a_g = 2, 3, 4$. With a positive g all contributions are positive. The cell input $\boldsymbol{z}$ (the function g) has a negative bias, that is, $\boldsymbol{b_z} < 0$. This is important to avoid the drift effect. The drift effect is that the memory content only gets positive contributions which lead to an increase of $\boldsymbol{c}$ over time. Typical values are $\boldsymbol{b_z} = -1, -2, -3, -4, -5$.

(A5) We want to ensure that $h(0) = 0$. If the memory content is zero then nothing is transferred to the next layer. Therefore we set $h = a_h \tanh$ with $a_h = 1, 2, 4$.

(A6) The memory cell content is initialized with zero at time $t = 0$, that is, $\boldsymbol{c}^0 = 0$.

The resulting LSTM forward pass rules are:

$$\boldsymbol{z}^t = a_g\, \sigma\left(\boldsymbol{W_z}\, \boldsymbol{x}^t + \boldsymbol{b_z}\right) \qquad \text{cell input} \tag{A358}$$

$$\boldsymbol{c}^t = \boldsymbol{z}^t + \boldsymbol{c}^{t-1} \qquad \text{cell state} \tag{A359}$$

$$\boldsymbol{y}^t = a_h \tanh\left(\boldsymbol{c}^t\right) \qquad \text{cell output} \tag{A360}$$

See Figure A14 for a LSTM memory cell without gates which perfectly distributes the relevance across the input.

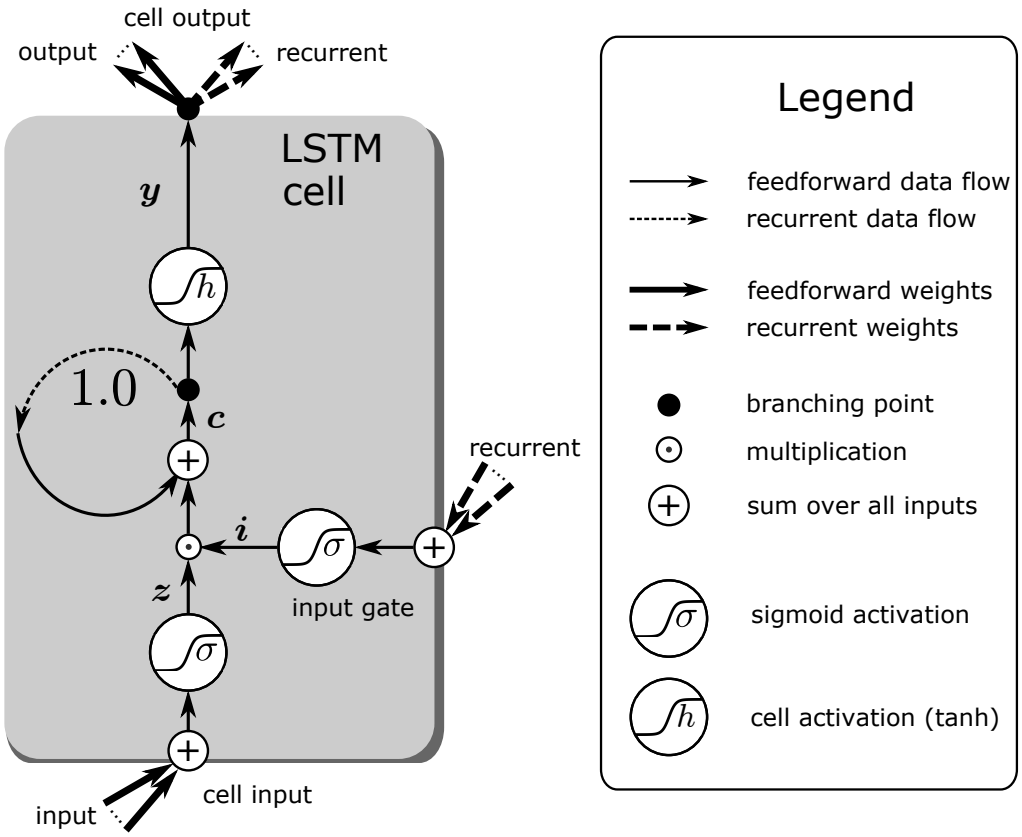

Figure A13: A nondecreasing LSTM memory cell.